

# **LAB - In class**

## **Topic 10: Review for final exam**



Lecturer: Lê Thị Nhàn, PhD.

Lab instructor: Nguyễn Ngọc Toàn, MSc.

Nguyễn Ngọc Minh Châu, MSc.

# Practice Questions for Final Exam

1. Find the names of all courses that have **at least one** prerequisite, but **none** of their prerequisite courses have any prerequisites themselves.
2. Identify **all** instructors who are teaching a course in the same semester and year as the department head of their own department.
3. Find all students who have taken and passed **at least one** course from every department in the university.
4. Find the department(s) with the **highest average salary** for their instructors.
5. Identify the students who are currently enrolled in a course but have not yet passed any of its prerequisite courses.
6. (CTE allowed) For each department, find the lecturer with the highest number of unique courses taught.
7. Identify the table of influence and write at least one trigger to ensure the BR1: There are no two courses within the same department having the same name.
8. Write a trigger to perform a **cascading delete** for a department.
9. Create a stored procedure to do the following steps with the given <student\_id> and <section\_id>:
  - a) Read the current grade of the student <student\_id> in the section <section\_id>.
  - b) Delay for 10 seconds
  - c) Reread the grade of that student
10. Create a stored procedure to delete a student grade given student\_ID and section\_ID.  
Input: student\_ID, section\_ID  
Output: Rollback if either student\_ID or section\_ID is non-exists. Otherwise, commit.
11. Create a stored procedure to multiply a student grade by 1.1 given student\_ID, section\_ID (new\_score = old\_score \* 1.1)  
Input: student\_ID, section\_ID  
Output: Commit if student\_ID and section\_ID exists. Otherwise, commit.
12. Use transactions to ensure that the stored procedures work as required. What would happen if two of the three SPs or all the three SPs are executed at the same time? What type of concurrency could happen at each isolation level? How would you handle it?